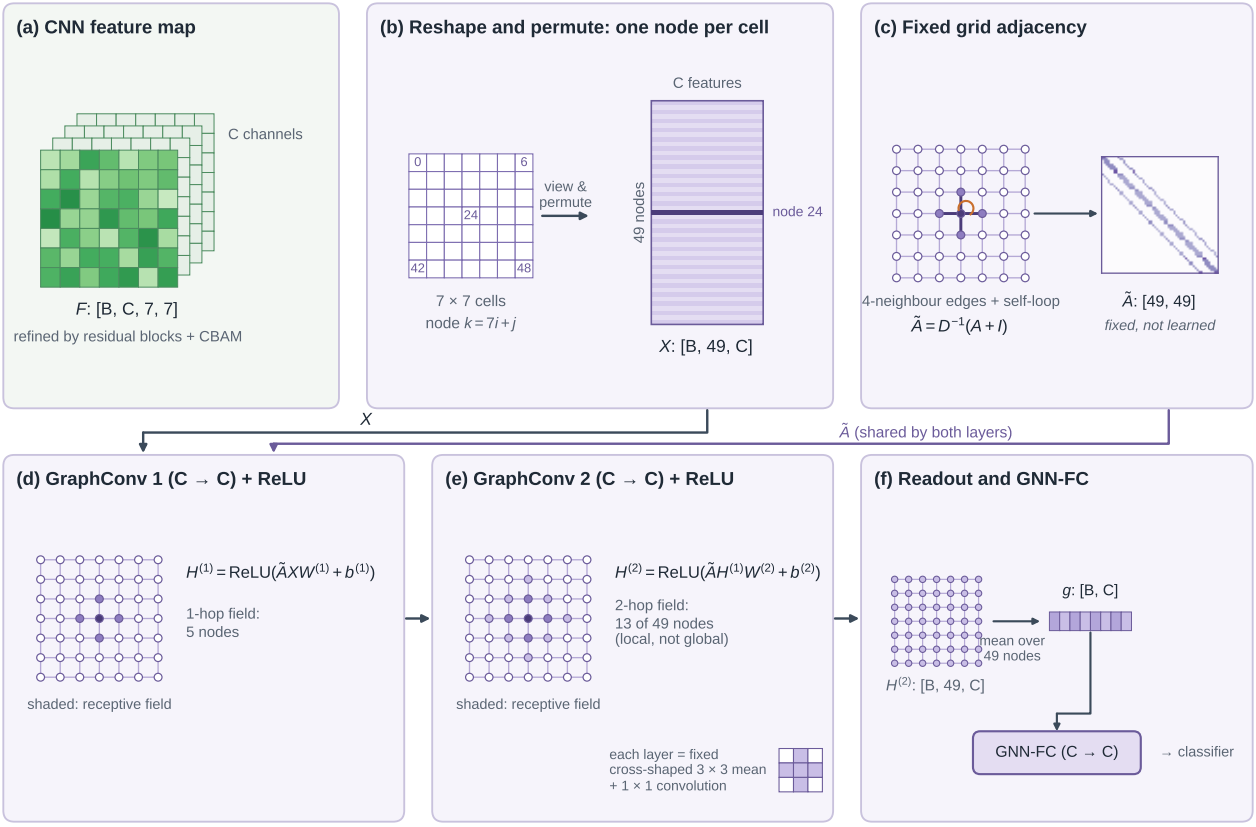

(a) CNN feature map
C channels
F: [B, C, 7, 7]
refined by residual blocks + CBAM
(b) Reshape and permute: one node per cell
0
6
24
42
48
7 × 7 cells
node k = 7i + j
view & permute
C features
49 nodes
node 24
X: [B, 49, C]
(c) Fixed grid adjacency
4-neighbour edges + self-loop
Ã = D⁻¹(A + I)
Ã: [49, 49]
fixed, not learned
X
Ã (shared by both layers)
(d) GraphConv 1 (C → C) + ReLU
H(1) = ReLU(ÃXW(1) + b(1))
1-hop field:
5 nodes
shaded: receptive field
(e) GraphConv 2 (C → C) + ReLU
H(2) = ReLU(ÃH(1)W(2) + b(2))
2-hop field:
13 of 49 nodes
(local, not global)
shaded: receptive field
each layer = fixed
cross-shaped 3 × 3 mean
+ 1 × 1 convolution
(f) Readout and GNN-FC
g: [B, C]
mean over
49 nodes
H(2): [B, 49, C]
GNN-FC (C → C)
→ classifier